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Structured representation and compression method of viewshed:
An example analysis of communication viewshed

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Abstract: Terrain viewshed analysis, which is a core spatial analysis technology, holds an important position in GISs. Viewsheds are typically represented as field models of discrete sets of visible scattered points in which the local clustering and spatial structural characteristics of the points are neglected. As a branch of terrain viewsheds, communication viewsheds are used to analyze signal reception with signal towers as observation points. In this study, a structured representation method in which communication viewsheds are used as examples, is proposed. First, each terrain point is treated as a base station, and the communication viewsheds are obtained. Second, each communication viewshed is segmented into subregions using feature recognition, and seven parameters are defined to describe the characteristics of each subregion. Finally, consistency and application verification are used to evaluate the effectiveness of the proposed methods. The experimental results show that the method can effectively reveal the spatial distribution characteristics of communication signal coverage, thereby providing theoretical support for quickly obtaining solutions to base station site planning problems. Furthermore, the structured representation method is also a lossy data compression method. This method is compared with the existing visibility data compression method, and the results indicate that the compression ratio can reach hundreds and increases exponentially with increasing terrain complexity.

Keywords: communication viewshed; structured representation; spatial structure characteristics; DEM; lossy compression

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1 Introduction

Since Miller first proposed the concept of digital elevation models in 1958, these models have undergone more than 60 years of development, and the basic theoretical problems associated with digital elevation models (DEMs) have been extensively studied (Xiong *et al*., 2021; Zhang *et al*., 2024). Viewshed analysis, which is a fundamental application of DEMs, has attracted significant attention (Tang, 2014) and is applied widely in site planning, landscape analysis and evaluation (San Martin Saldias and McGlade, 2023), spatial cognition and decision-making, archaeology, military operations, and other fields (Zhang *et al*., 2005). The concept of a communication viewshed, which was introduced by Beak and Choi in 2018, has since emerged as a specialized branch of viewshed analysis. A communication viewshed describes the signal reception at terrain points when the observation point is a base station and effectively represents signal coverage (Beak and Choi, 2018).

Communication viewsheds are also used widely in industrial circles. A classic application is base station location planning, and intelligent algorithms, such as genetic algorithms (Yang *et al*., 2022), particle swarm optimization algorithms (Huang *et al*., 2022), and simulated annealing algorithms (Zhang *et al*., 2022), are typically used to solve this problem. With the continuous development of UAV technology, in remote mountainous areas, the sea, and other areas where signal coverage is not available, a UAV can be equipped with communication equipment as an aerial base station to provide communication network services (Li and Xing, 2018). In addition, during a large event, UAVs can be deployed over an event site to provide additional communication capacity to meet the communication needs of many people. In the military, UAVs can be used as communication nodes to ensure smooth communication between combat troops (Gargalakos, 2021).

With the gradual popularization of 5G networks, research in the field of communication viewsheds has become urgent, and the calculation of communication viewsheds is among the main research topics. The research on and development of terrain visibility analyses on the basis of DEMs have led to the accumulation of many research results, and research on single-point viewshed calculation methods, such as the R3 algorithm (Van Kreveld, 1996), Xdraw algorithm, reference plane algorithm (Wang *et al*., 2000; Wu *et al*., 2007), HiXdraw algorithm (Zhu *et al*., 2019), PDERL algorithm (Wu *et al*., 2021), and DPDERL algorithm (Tang *et al*., 2024), is relatively mature. These algorithms can effectively improve the accuracy and efficiency of single-point-viewshed calculations. However, the concept of a communication viewshed was only recently proposed by scholars, and relatively few studies have been conducted on the direction of communication viewsheds. Before communication viewsheds were proposed, researchers proposed corresponding signal propagation models to calculate the signal coverage of base stations for specific application scenarios and natural environments. For example, a modified propagation model of terahertz radio was proposed for indoor communication scenarios (Peng *et al*., 2022). Zhang *et al*. (2020) used the logarithmic distance attenuation model to fit the signal receiving intensity in five forest environments, and Chen *et al*. (2021) provided the range and conditions applicable to the logarithmic loss model in the Masson pine forest environment. However, most of these models are not universally applicable. In 2018, for the first time, Beak and Choi proposed a communication viewshed and a calculation method for the communication viewshed in the 3D Fresnel region (Beak and Choi, 2018). However, this method does not fully consider the

diffraction characteristics during signal propagation. Additional improvements were achieved using the communication viewshed computation algorithm for wireless signals (CVCWS) algorithm, which was proposed by Wang and Dou (2024) to improve the accuracy of the computation of communication viewsheds.

For the calculation of a communication viewshed or its related applications, the communication viewshed is usually simplified to a discrete point set, and local agglomeration and overall spatial structure are ignored. This approach not only increases data redundancy but also limits the in-depth understanding of signal coverage characteristics. The analysis and mining of the abundant geographic information contained in massive amounts of data have become hot research issues. Ling *et al*. (2023) focused on data in the field of natural resource management and organized and expressed the multidomain geographic entity information contained in text through a quadruple of semantic description, spatial location, attribute characteristics, and time evolution. Zhu *et al*. (2024) analyzed the elemental compositions and logical structures of terrain scenes and used semantic elements with time, place, things, events, and phenomena as the core elements to describe them. Although various methods have been proposed for organizing and expressing the geographic information in large amounts of text data in the field of geoscience, only Zhang *et al*. (2025) investigated the spatial distribution structure of viewsheds. They segmented the viewsheds using the density-based spatial clustering of applications with noise (DBSCAN) algorithm. However, they did not elaborate on the accuracy of the segmentation; they only studied the spatial distribution of binary viewsheds without forming a complete structured representation. This research is at the initial exploration stage and is thus not mature. The distribution of a viewshed is relatively independent and dispersed, and the viewshed data are abstracted into a discrete point set, which makes the rich structural information scattered, highly unstructured, and lacking effective organizational expression. The representation of communication viewshed data is consistent with the traditional viewshed data representation method, which lacks structural information, thereby resulting in a large data volume, low analysis efficiency, and high optimization difficulty in large-scale base station location planning. The study of structured representation method that can efficiently and accurately express the features of a communication viewshed is of great theoretical importance and practical value. Therefore, in this study, a structured representation method for communication viewsheds that is based on DEMs and organizes and expresses geographic information contained in communication viewshed data through structured description is proposed.

Furthermore, with the rapid development of remote sensing technology, the scale of terrain data is increasing, which is leading to the need for more storage space for terrain viewshed data. Therefore, an effective data compression method is needed. Research on terrain data compression methods is relatively advanced. Terrain data compression methods that are based on DEMs include mainly lossless compression methods and lossy compression methods. The lossless compression methods include the SPIHT algorithm (Said and Pearlman, 1996), the JPEG2000 standard (Christopoulos *et al*., 2000), arithmetic coding (Kim and Wen, 2007), and the DCR algorithm (Son *et al*., 2011), whereas the lossy compression methods include height-domain compression (Gerstner, 2003) and compression on the basis of the wavelet transform (Wang and Zhu, 1999). However, relatively few compression methods for viewshed data that are based on DEMs exist. Mineter *et al*. (2003) gener-

ated a complete intervisibility database (CID) to save the visibility data of each position in a DEM and performed lossless compression on the visibility data with a compression ratio of approximately 33. In large-scale terrain viewshed application scenarios, the compression efficiency still fails to satisfy the actual storage and transmission requirements. To address this issue, in this paper, a lossy compression method that is based on a communication viewshed structured representation is proposed, which significantly improves the compression ratio and is also applicable to terrain-viewshed data.

To the best of our knowledge, previous studies have not deeply examined structured representation methods for communication viewsheds. This study fills the gap in this direction with three innovations:

(1) A structured representation method for communication viewsheds is proposed, which fully considers the spatial aggregation of a communication viewshed, describes the spatial distribution of base station coverage, and provides methodological support for base station location planning.

(2) The proposed structured representation method, which is based on communication viewsheds, is a lossy compression method. Converting communication-viewshed data into structured representations for storage can significantly reduce the occupation of storage space. Compared with existing methods, the compression ratio is significantly improved, thereby effectively solving the problems of data storage and transmission in large-scale base station location applications.

The remainder of this paper is organized as follows. In the first chapter, the Introduction, a summary of the research status, and proposed questions are presented. In the second chapter, the overall framework of the proposed structured representation method for communication viewsheds is presented. In the third chapter, a feature extraction method for the communication viewshed of a single base station is described. In the fourth chapter, a validity evaluation and effect analysis of the communication viewshed structured representation and a comparative experiment on data compression methods are presented. In the fifth chapter, a summary of this study is provided.

2 Overall framework of the structured representation method for communication viewsheds

The overall framework of the structured representation method for communication viewsheds that is proposed in this study is illustrated in Figure 1. It consists of 4 steps.

1) Calculation of the communication viewshed

Each point on the terrain is regarded as the location of a base station. In this study, the CVCWS algorithm is used to calculate the communication viewshed. However, because many terrain points need to be calculated from the communication viewshed of the base station, parallel computing is used to increase the calculation efficiency.

2) Extraction and fitting of communication viewshed aggregation areas

The communication viewshed is spatially aggregated. After each aggregation area is extracted using image processing technology, to obtain a more structured representation, each aggregation area is fitted with an approximate shape.

3) Setting of the fitting area parameters

To achieve a more accurate structured representation and effectively reflect the infor-

mation in the original communication viewshed data, seven parameters are set for each fitted aggregation area: region size, shape, center coordinates, density, communication quality, the standard deviation of communication quality, and the key coordinate points of the enclosed area.

4) Structured representation of a single-base-station communication viewshed

Because the parameters in Step 3 alone cannot form a complete semantic, a structured description template is designed to organize the parameters and form a structured representation, thereby providing theoretical and technical support for subsequent application research on base station location planning.

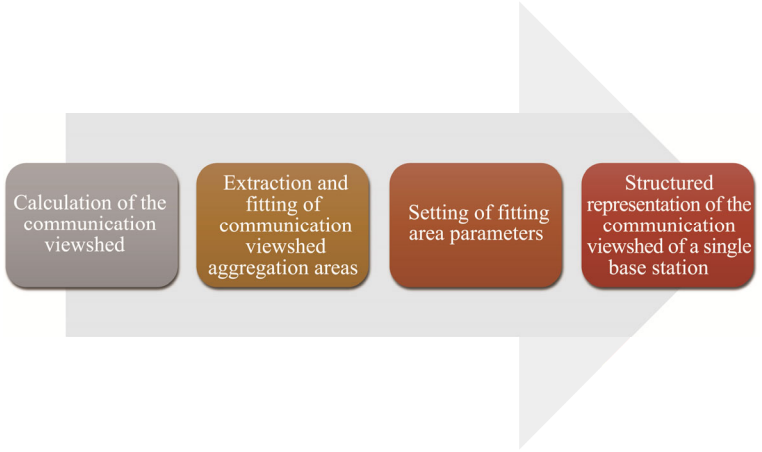


Figure 1 Overall framework of the communication viewshed structured representation method

3 Structured representation method for the communication viewshed of a single base station

3.1 Obtaining the communication viewshed

Extracting the structural information of the communication viewshed requires calculating all the terrain points on the given terrain as the communication viewshed of the base station in advance. In this study, the CVCWS algorithm was used to calculate the communication visibility quality of base stations at different positions, and these results were taken as the communication viewshed. Because the CVCWS algorithm is sensitive to the size of the terrain data, the calculation time increases exponentially with increasing data volume. Calculating the communication viewshed of a single base station on a 400×400 terrain took 15 min. For a terrain of size 800×800, the computing time for a single base station increased to 218 min. Because each terrain point needed to be obtained as the communication viewshed of the base station in this study, the number of computations was large. Therefore, the CUDA parallelized computing method was adopted to reduce computing time and improve efficiency.

The communication viewshed calculated in this study reflects the signal reception quality at each terrain point under the current base station, and its value was normalized to the range [0, 1]. Communication viewshed diagrams for the base station at three different terrain points are shown in Figure 2.

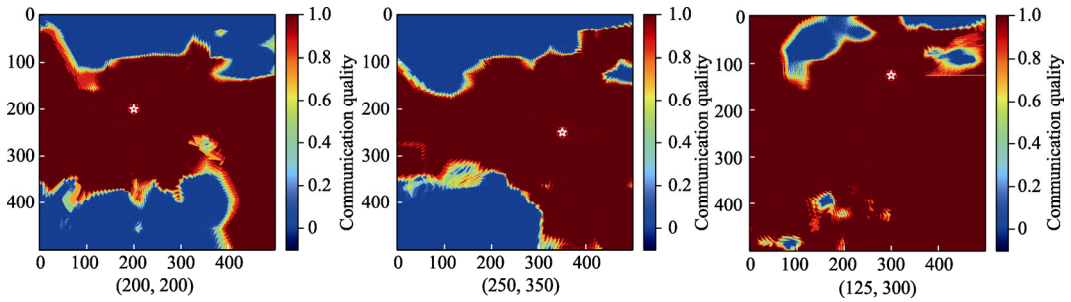


Figure 2 Schematic diagrams of the terrain and communication viewshed for the base station at three different terrain points (the red part of each figure indicates that the communication quality is close to 1 and is good, whereas the blue part indicates that the communication quality is close to 0 and is poor.)

3.2 Region extraction and fitting

As shown in Figure 2, the quality of communication in the communication viewshed is spatially clustered. The clustering method can identify and separate spatially clustered regions. The density-based spatial clustering of applications with noise (DBSCAN) algorithm and ordering points to identify the clustering structure (OPTICS) algorithm are both density-based clustering algorithms with obvious advantages in processing spatial data. Both DBSCAN and OPTICS clustering involve the selection of parameters that significantly affect the clustering results. The DBSCAN and OPTICS algorithms have low efficiency in large-scale data clustering; therefore, they are not suitable for the division of spatially clustered regions in the communication viewshed of any terrain or base station. In this study, the communication viewshed was expressed as an image, and feature extraction technology from image processing was used to effectively extract spatial aggregation areas.

To facilitate the extraction feature of the communication viewshed, the communication quality values were first dichotomized into two classes: good and poor. This classification required defining a threshold *T*. The choice of *T* was critical because it directly determined the spatial extent of the regions to be extracted. In this study, the communication viewshed of the base station located at (200, 200) in Figure 2 was selected for sensitivity analysis. The threshold T was varied from 0.1 to 0.9 in steps of 0.1. The variation in the communication quality of the largest region is shown in Figure 3. As shown in Figure 3, when the threshold T was less than 0.7, the communication quality gradually improved. When the threshold T was greater than 0.7, the communication quality remained essentially unchanged. Therefore, in this study, the threshold T was chosen to be 0.7. A communication quality value greater than or equal to the threshold in the communication viewshed was considered to indicate good communication, and the corresponding value was set to 1. A communication quality less than the threshold was considered to indicate poor communication, and the corresponding value was set to 0. A binary image was formed, which laid the foundation for subsequent feature extraction.

The quality of communication in the communication viewshed after division according to the threshold value has a more obvious boundary, as shown in Figure 4. In Figure 4b, if all black areas can be extracted, the remaining white areas can be considered as a whole. The black areas indicate good communication quality, whereas the white areas indicate poor

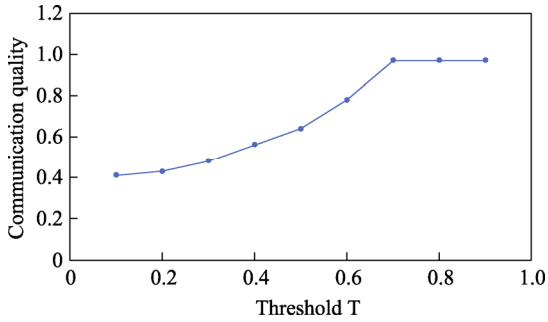


Figure 3 Sensitivity of regional communication quality to the binarization threshold T

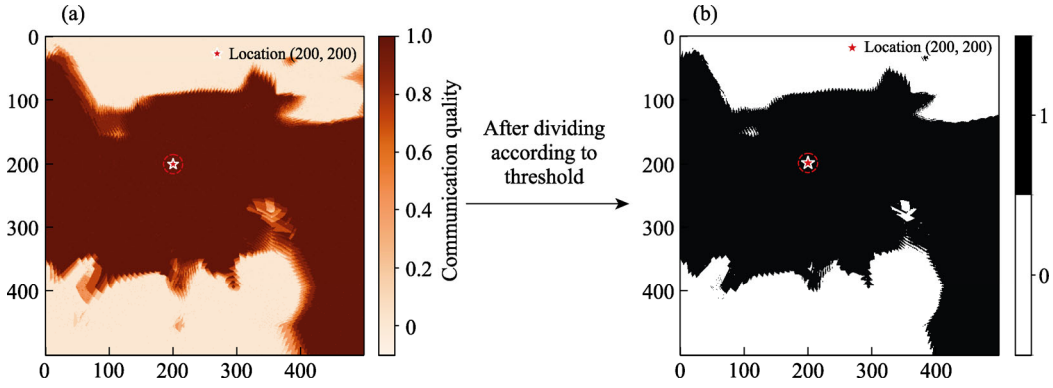


Figure 4 Comparison of the communication viewshed before and after threshold division (the black area in (b) indicates good communication quality, whereas the white area indicates poor communication quality.)

communication quality. However, the communication viewshed calculated on different terrains still has problems, such as poor communication quality, gaps in the region, and blurred boundaries in the spatial aggregation region (as shown in Figure 5). Because of these gaps, a single, contiguous region of interest may appear as multiple disconnected components in the binary image, which would lead to over segmentation during feature extraction. To better identify the contours of high-aggregation and high-density regions and extract their spatial features, the following morphological operations were adopted: closing and expansion. The closing operation can fill the small gaps in the region and increase the integrity of the region to effectively avoid the problem of incorrect segmentation of the target region caused by noise. The boundary of the target region can be extended by the expansion operation, which makes the edge clearer and reduces the difficulty of the subsequent contour detection. Finally, the feature extraction method was used to identify the contours of all the connected target regions in the binary image. The feature extraction method not only has high recognition accuracy but can also quickly realize large-scale data processing and feature extraction and meet real-time requirements in practical applications.

The use of contours to represent a region requires the storage of many boundary points, which requires more storage space. Moreover, contours cannot concisely describe the shape of the region. Each contour was fitted into an approximate graph, which was selected from among five types of fitting graphs: concave, convex, rectangular, oval, and triangular. To quantitatively select the best-fitting shape from the five candidate types, a mathematical

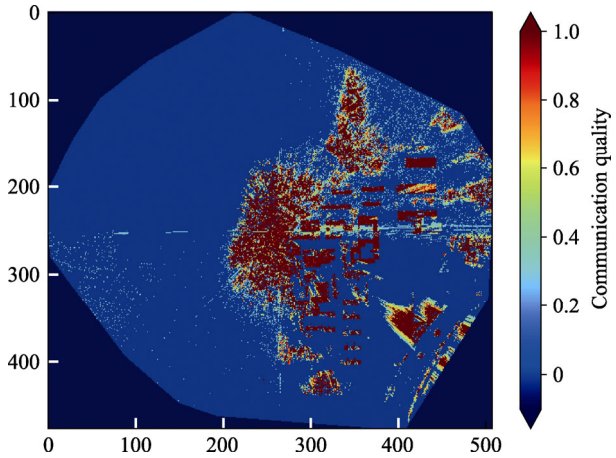


Figure 5 Schematic diagram of gaps in the communication viewshed (the fuzzy boundaries of the spatially aggregated regions in this communication viewshed lead to a reduction in the accuracy of feature extraction.)

error metric was defined. For a given contour C and a candidate fitted shape S, the fitting error $E(C, S)$ was calculated using Eq. (1):

$$E(C,S) = \frac{|A_C - A_S|}{A_C} \tag{1}$$

where A_C represents the area enclosed by the original contour C and A_S represents the area of the candidate fitted shape S. The shape S that minimizes this error metric was selected as the best approximation. The calculation formula of S^* is shown in Eq. (2):

$$S^* = \arg \min_{S \in \{\text{concave, convex, rectangle, oval, triangle}\}} E(C,S) \tag{2}$$

This metric ensures that the selected shape best preserves the spatial extent of the original aggregated region while allowing for geometric simplification.

The region was identified using the feature extraction method and approximated using a fitting graph, which provided the basis for subsequent region-feature analysis and structured representation.

3.3 Setting the region parameters

In Step 1, the communication viewshed was divided into several subregions. To describe the characteristics of each subregion more accurately, the following seven parameters were defined:

(1) Region_density: The degree of aggregation of terrain points with good communication quality in the region can be measured by dividing the number of points with good communication quality in the fitting area by the number of terrain points in the fitting area, as shown in Eq. (3):

$$region_density = n / N \tag{3}$$

where n represents the number of points with good communication quality in the fitting region and N represents the total number of points in the fitting region.

(2) Region_quality: The communication quality in the region can be calculated by dividing the sum of the signal reception quality values of all the terrain points by the number of

terrain points in the region, as shown in Eq. (4):

$$region_quality = \sum_{N}^{1} S_i / N \tag{4}$$

where S_i denotes the communication quality value of the i-th terrain point.

(3) Region_quality_std: To describe the heterogeneity of the signal strength within the region, the standard deviation of the communication quality can be calculated as shown in Eq. (5):

$$region_quality_std = \sqrt{\frac{\sum_{i=1}^{N}\left(S_i - region_quality\right)^2}{N}} \tag{5}$$

A low *region_quality_std* indicates uniform signal coverage within the region, whereas a high value suggests significant variation, which is critical for assessing coverage stability.

(4) Region_area: This quantity represents the fitting area and is obtained by directly calculating the area of the fitting shape.

(5) Region center coordinates (x,y): These coordinates represent the center coordinate point of the fitted shape.

(6) Region_shape: From among five fitting shapes, namely, an oval, a rectangle, a triangle, a convex polygon, and a concave polygon, the shape with the smallest approximation error is selected.

(7) Region_keypoints: All the key coordinate points that enclose the fit shape are listed to reflect the fit region.

These parameters can be used to describe regional characteristics more accurately.

3.4 Structured representation of a single base station

In Step 2, seven parameters were defined for each region, but the parameters alone could not constitute a complete semantic representation. To improve the comprehensibility of the data, seven parameters that corresponded to each fitting region were filled into the structured description template to obtain the structured representation result for the communication viewshed of a certain base station. The structured description template of the communication viewshed of a single base station is shown in Figure 6. In Figure 6, the red font represents

The communication viewshed at the (*Terrain Coordinates*) position can be divided into (*N*) zones in total.

The (*1*) fitted area can be approximated by a (*region_shape*) graph, with an approximate area of (*region_area*) m², a center coordinate of (*x, y*), a point density of (*region_density*), an average communication strength of (*region_quality*), a standard deviation of signal strength of (*region_quality_std*) and region key points formed by (*region_keypoints*).

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The (*N*) fitted area can be approximated by a (*region_shape*) graph, with an approximate area of (*region_area*) m², a center coordinate of (*x, y*), a point density of (*region_density*), an average communication strength of (*region_quality*), a standard deviation of signal strength of (*region_quality_std*) and region key points formed by (*region_keypoints*).

Figure 6 Structured representation template of the communication viewshed of a single base station

the parameters defined in Step 2.

Taking each point on the terrain as the location of the base station to calculate its communication viewshed and translating the communication viewshed of each base station according to the above three steps is not only an effective data compression method that can significantly reduce the storage requirements of communication-viewed data but also reflects the spatial distribution characteristics of the signal coverage of each base station. This approach provides technical and theoretical support for solving the problem of base station location in practical applications.

4 Method validation

To validate the proposed communication viewshed structured representation method, two experiments were designed for consistency and application verification. The effect of the communication viewshed structured representation method was experimentally analyzed. In addition, the accuracy of the structured representation as a lossy data compression method was analyzed through experiments. By comparing the structured representation compression method in this study with existing viewshed compression methods, the advantages of this structured representation compression method were explored.

4.1 Effectiveness evaluation

4.1.1 Verification of consistency

A consistency verification experiment can evaluate whether a structured representation can accurately describe the original data. The structured representation method that was used in this study involved dividing the communication viewshed into several subregions and then describing each subregion using structured representation combined with parameters. If the original communication viewshed and the graph translated according to the structured representation convey the same information, the proposed communication viewshed structured representation method is effective.

The design concept for the consistency verification experiment was as follows: First, the communication viewshed of a single base station was described using the structured representation method proposed in this study. Second, the structured representation results were automatically analyzed to form a corresponding diagram on the basis of the analyzed content. Because the coverage radius of most base stations in real life ranges from 100 m to 1 km, a terrain with a size of 500×500 and a resolution of 1 m was selected for the experiments, as shown in Figure 7. All the points on the terrain were taken as base stations, with the height of the base stations assumed to be 10 m and the signal-receiving height assumed to be 2 m. The parallel calculation method proposed in Section 2 was used to calculate the communication viewsheds of all the base stations. The communication viewshed results for three different base stations were randomly selected, and a feature extraction method was used for structured representation. The locations of the three base stations were (125, 300), (250, 350), and (400, 100). The communication viewsheds of the three base stations are shown in Figure 8. When feature extraction was carried out, the regional division of the communication viewshed of each base station was as shown in Figure 9.

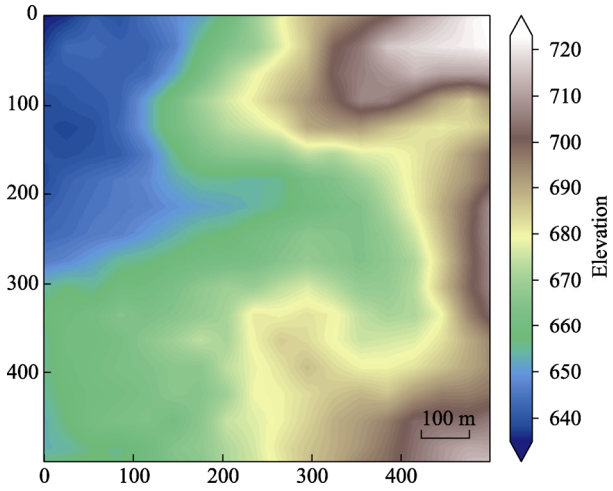


Figure 7 Terrain diagram for the experiment

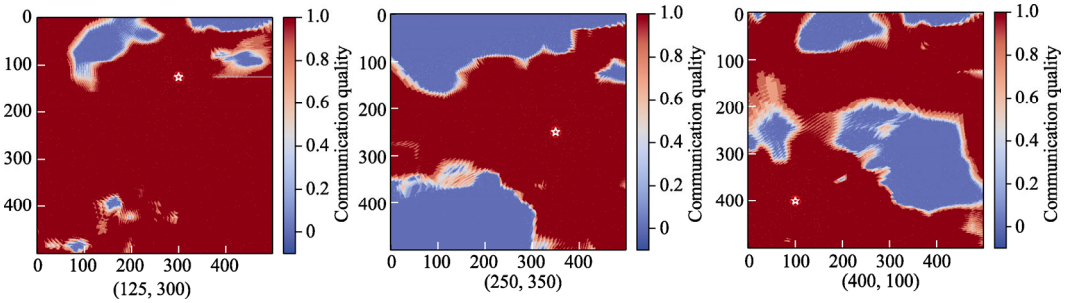


Figure 8 Schematic diagrams of the communication viewsheds of base stations at three terrain points

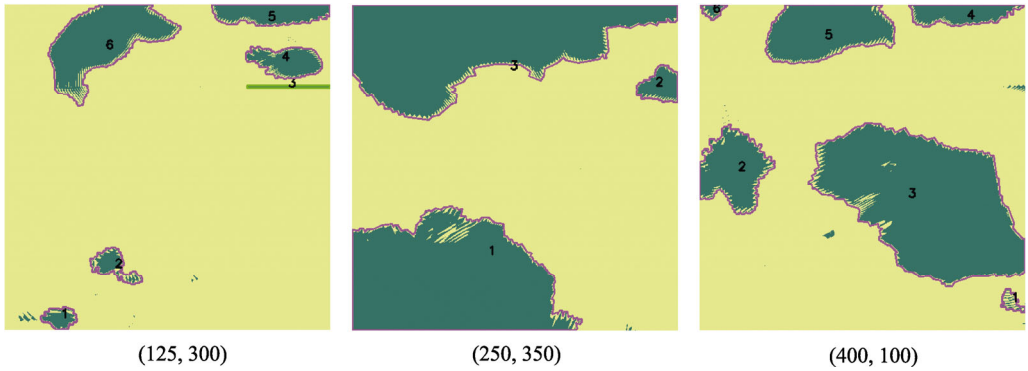


Figure 9 Schematic diagrams of the regional division of the communication viewsheds of three base stations during feature extraction

The results of analyzing the structured representation of the communication viewshed of the three base stations and translating them into graphs are shown in Figure 10. In Figure 10, the spatial distribution characteristics of the signal coverage of the base stations and changes in the signal strength can be visually observed. Graphs were formed using structured representation analysis and translation. Therefore, the feature extraction and expression method proposed in this study can be indirectly proven to effectively describe the coverage of the

communication viewshed of the base station. A comparison of Figure 10 with Figures 8 and 9 reveals that the structured representation translated into an image is similar to the information transmitted by the original communication viewshed image and can also show the signal strength of each subregion, which verifies the accuracy of the feature extraction and expression method from this perspective.

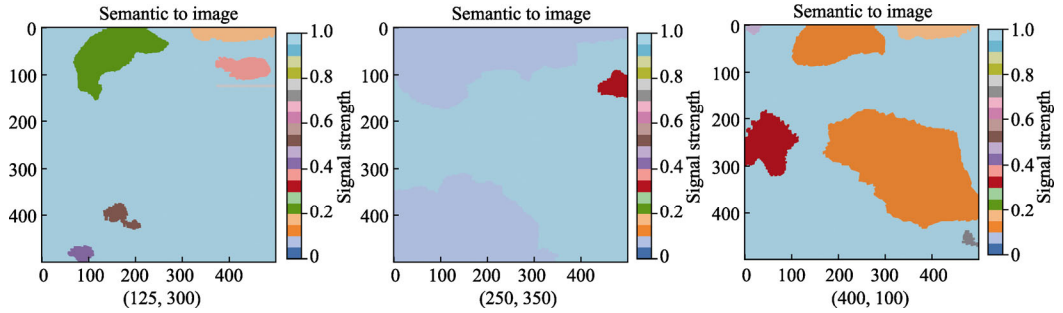


Figure 10 Results of analyzing the structured representations of the communication viewsheds of the three base stations and translating the analytic content into graphs

4.1.2 Application verification

In addition to the consistency verification experiment, in this study, an application verification experiment was designed to evaluate the effect and feasibility of the proposed structured representation method in practical applications by applying it to specific problems in actual scenarios. Point cloud data for the northern terrain of the Xianlin Campus of Nanjing Normal University were obtained from the National Earth System Science Data Center (http://geodata.nnu.edu.cn). On the basis of the point cloud data, DEMs data with an accuracy of 2m were generated, as shown in Figure 11a. Two areas with dense buildings and populations were selected as demand areas for signal coverage on this terrain, as shown in Figure 11b. Choosing a location on this terrain to construct a base station would improve communication between these two areas.

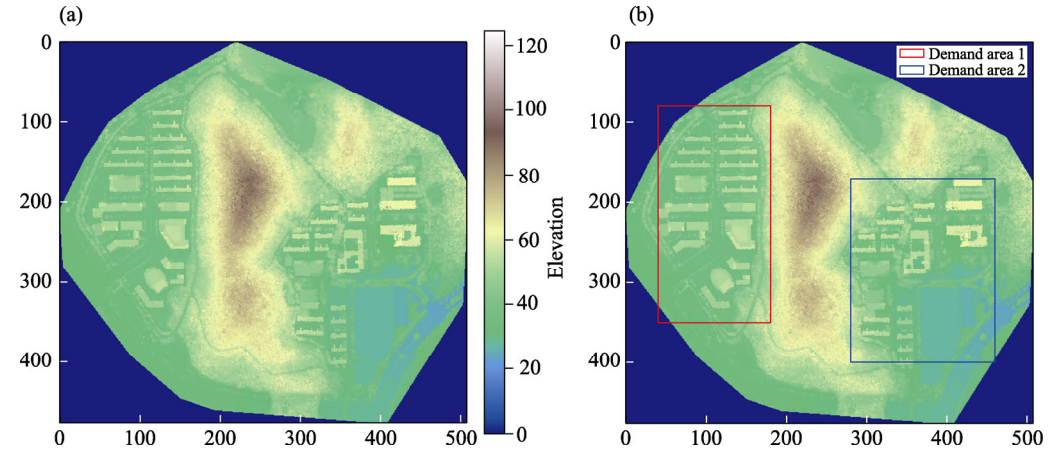


Figure 11 Topographic map of the North District of Nanjing Normal University (a) and signal coverage demand area (b)

First, the communication viewsheds of all the terrain points as base stations were calculated and structured. On the basis of the structured representation results for each base station, it was matched with the area to be covered. When matching, three indicators—coverage, coverage communication quality, and coverage density—were used to calculate the matching degrees of the base stations at different locations, and these three indicators were weighted to obtain a matching score for each base station. The matching degree calculation equation is given by Eq. (6):

$$matching_\deg ree = w_1 \times rate + w_2 \times quality + w_3 \times density \tag{6}$$

where *rate* represents the coverage rate, *quality* represents the communication quality of coverage, *density* represents the coverage density, and w_1, w_2, and w_3 represent the corresponding parameters of the three indices.

In this experiment, w_1, w_2, and w_3 were set to 1/3, and the degree of matching under the current coverage requirement was calculated on the basis of the structured information of each base station. The communication viewsheds of the top four base stations in terms of matching degree score and the region division during the extraction and expression of the corresponding structured information are shown in Figure 12. The coordinates of these four base stations on the terrain were (186, 238), (172, 235), (331, 225), and (311, 220), respectively, among which the base station located at (186, 238) had the highest matching score. According to the communication viewshed map, the base station in this position had the best coverage effect on the demand area and could effectively meet the signal coverage requirements of the target area. Table 1 shows the index values (coverage, coverage density, and coverage intensity) and matching degree values for the top four base stations with the highest matching degrees. Table 1 also shows that the base station at position (186, 238) had relatively high scores for various indicators in the matching process. Its comprehensive matching degree was higher, thus demonstrating its advantages in terms of coverage of the demand area. Although w_1, w_2, and w_3 are all set to 1/3 in this experiment, the weight values of w_1, w_2, and w_3 can be adjusted flexibly according to changes in specific coverage requirements.

To demonstrate the advantages and effectiveness of the structured representation method, an efficiency comparison experiment was conducted to solve the single-base-station location problem on the basis of the terrain shown in Figure 11a and the demand area shown in Figure 11b. The genetic algorithm is a common method for solving a single-base-station location problem. Using this algorithm, in this study, each individual (solution) was encoded as the (x, y) coordinates of a candidate base station location. An initial population of 50 individuals was generated uniformly at random across the entire terrain. The fitness of each individual was calculated as the *matching_degree* via Eq. (6), i.e., the weighted sum of the coverage rate, coverage quality, and coverage density with respect to the demand areas. This ensured that both methods optimized the same objective. Tournament selection with a tournament size of 3 was used. A two-point crossover operator was applied with a probability of P_c=0.8. A Gaussian mutation operator was applied to each gene with a probability of P_m=0.05. The algorithm terminated after reaching 200 generations or if the best fitness did not improve for 50 consecutive generations, whichever occurred first. The solution times of the demand area matching method that is based on structured representation and the genetic algorithm and those of other algorithms are listed in Table 2. As shown in Table 2, the

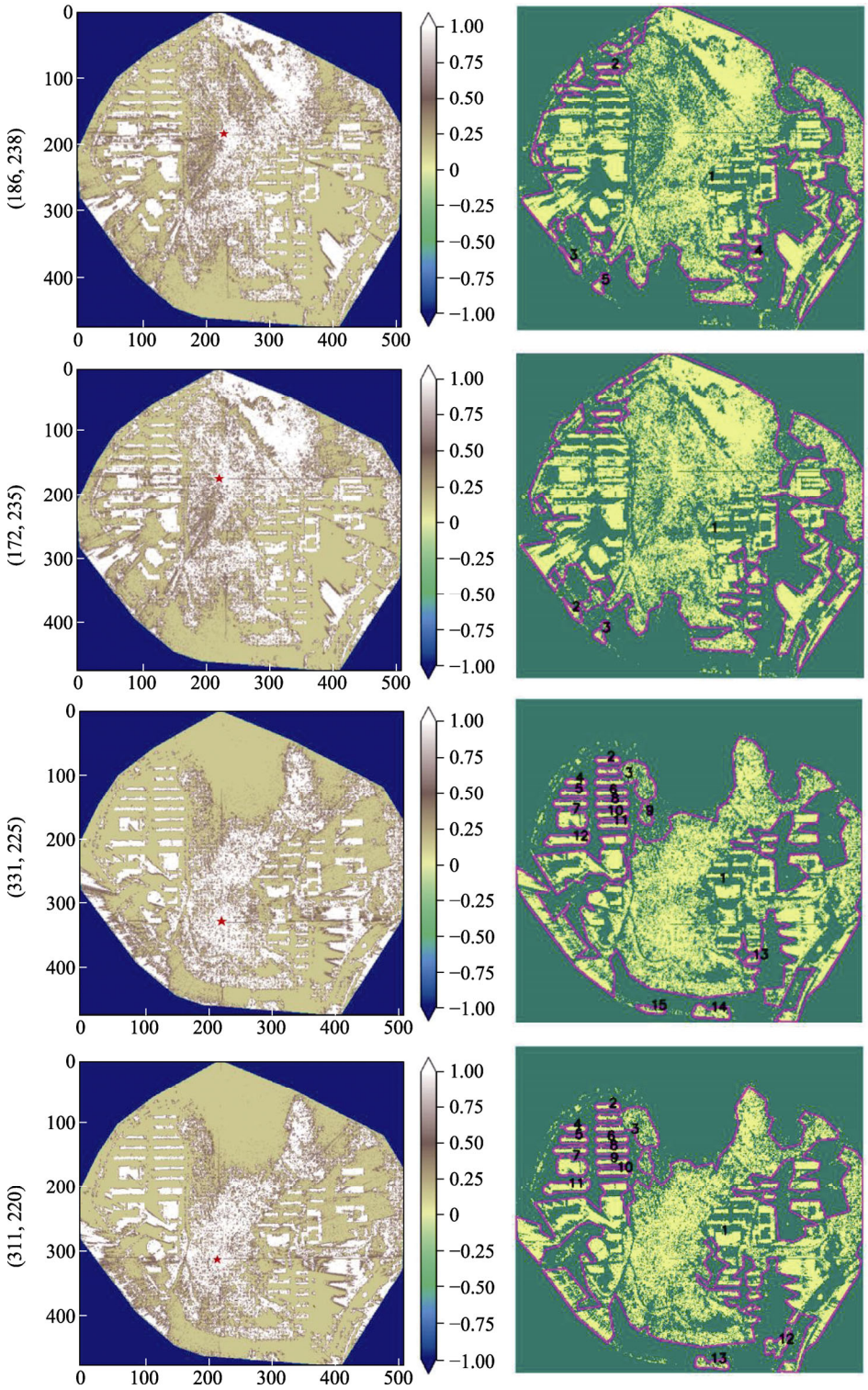


Figure 12 Schematic diagrams of the communication viewsheds of the top four base stations in terms of the matching score ranking and the regional division results obtained via feature extraction (the color bar in the figure represents the communication quality.)

Table 1 Index values and matching degree values of the top four base stations in terms of matching scores

Base station coordinates	Coverage rate	Coverage density	Coverage strength	Degree of matching
(172, 235)	0.762	0.457	0.94	0.974
(186, 238)	0.769	0.441	0.962	0.981
(311, 220)	0.634	0.438	0.991	0.899
(331, 225)	0.665	0.433	0.986	0.916

Table 2 Computation times of two algorithms for solving a single-base-station location problem

Requirement region matching method based on structured representation (s)	Genetic algorithm (s)
12	2430

calculation time for solving the single-base-station location problem using the demand area matching method proposed in this study was the shortest, and the efficiency was the highest. These findings were attributable to the structured representation containing the base station coverage distribution information, the signal coverage quality being aggregated, and the signal coverage quality within the same area being similar. Therefore, after the structured information were parsed and demand area matching was performed, the best location result was quickly obtained. In contrast, the genetic algorithm needs to search and evaluate the solutions in the entire population during the solution process, with point-by-point judgments between the demand area and the communication viewshed constantly performed. The calculation time is significantly affected by the population size and number of iterations; thus, it takes longer and has a lower efficiency.

4.1.3 Analysis of the advantages of the communication-viewshed structured representation

The base station was set at coordinates (200, 200) on the terrain in Section 4.1.1. The spatial distribution and coverage of the communication viewshed, as reflected by the parameters in the structured representation, are shown in Figure 13. The signal coverage of this base station can be divided into four areas, and its parameter values are listed in Figure 13.

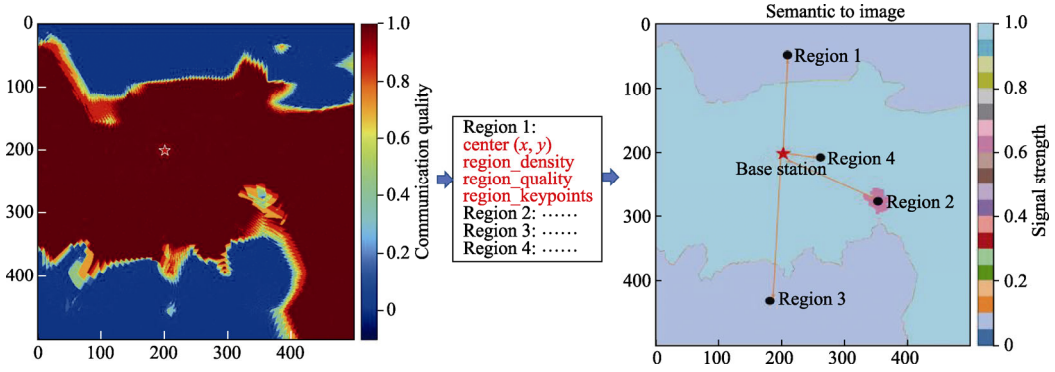


Figure 13 Comparison of the spatial distribution of the communication viewshed as reflected by the parameters in the structured representation

Because there were multiple keypoints enclosing the area, in Table 3, under Region_keypoints, only one of the keypoints is listed, and the remaining keypoints are replaced by ellipses. Traditional base station communication viewshed data are usually stored in text

form, where the signal reception status at each terrain point is represented discretely. The method of representing data by discrete points cannot reflect the spatial aggregation or distributional characteristics of signal reception. However, after structured expressing the communication viewshed data, the spatial structure of the communication viewshed can be clearly reflected through three parameters, namely, the coordinates of the regional center point, the fitted shape, and the key coordinate points, thereby enabling an understanding of the spatial distribution of the base station. Two parameters, namely, region_density and region_quality, can reflect the signal coverage intensity of different areas of the base station. Therefore, the structured representation of the communication viewshed can effectively organize the information represented in discrete data.

Table 3 Parameters of each area covered by the base station at (200, 200)

Region	Region_ density	Region_ quality	Region_ quality_std	Region_ area	Center coordinates	Region_ shape	Region_ key-points
1	0.99	0.08	0.077	49155.5	(227.86, 85.23)	concave_poly	(0, 37)…
2	1.00	0.60	0.054	929.5	(348.16, 276.91)	concave_poly	(328, 267)…
3	0.99	0.09	0.063	55846.5	(227.96, 371.18)	concave_poly	(414, 390)…
4	0.99	0.96	0.071	144068.5	(250.12, 209.58)	concave_poly	(499, 127)…

The proposed method performs well in contiguous, high-density coverage areas. However, a notable limitation emerges when dealing with regions of low regional density, where "good" signal points are sparse and scattered. In such cases, the initial binary image contains multiple isolated areas. Although morphological dilation and closing operations are effective at filling small gaps, they tend to overmerge these distinct clusters. During the extraction stage, several small, discrete coverage patches are connected into a larger contour. This erroneous merging adversely affects the structured representation. First, the fitted shape for this over merged contour inevitably encompasses areas that actually have poor or no signal coverage. Second, the calculated *region_quality* for the larger region averages the strong signals from the original small clusters with the zero-signal areas now included in the contour, thereby reducing the accuracy of the structured expression.

4.2 Performance evaluation of the structured representation compression method

4.2.1 Accuracy analysis of the structured representation compression method

In Section 4.1, the effectiveness of the structured representation method proposed in this study was demonstrated through two experiments. In this section, the accuracy and advantages of the structured representation compression method are analyzed. When the communication viewshed of each base station was structured expressed, the region was divided by extracting the contour, and each contour-enclosed region was approximated using a fitted shape. During the fitting process, errors were incurred, which affected the results of the structured representation compression method. However, because the best shape-fitting method was adopted in this study, five common fitting shapes were established: ellipses, rectangles, triangles, convex polygons, and concave polygons. When each contour region was fit, the shape with the smallest error among the five shapes was selected for approximation, thereby effectively reducing the fitting error. Moreover, experiments have revealed that among the above five shapes, the error is significant only when a concave polygon is used

for approximation. In this study, the Ramer–Douglas–Peucker algorithm was used to simplify and approximate the contour and control the number of boundary points retained by setting a threshold ε. With a smaller ε value, more boundary points are retained, which results in a smaller fitting error and greater precision. With a large ε value, details can be ignored, thereby significantly simplifying the fitting shape to a greater extent. Therefore, by setting different ε values, the fitting shape can be adapted to different application scenarios with different requirements. The ε value was set to 1 meter (equivalent to 1 pixel given the 1 m resolution of the DEM) in this study, which was aligned with the spatial resolution of the terrain data. This choice ensured the preservation of macrostructural features by filtering sub-resolution noise while maintaining key inflection points and achieved an optimal balance between shape fidelity and data compression. The fitting errors of each region when structured expressing the communication viewshed of the base stations at three different locations are listed in Table 4. The error was calculated using the three different points (125, 300), (250, 350), and (400, 100) on the terrain in Section 4.1.1, and the accuracy of the fitted shape was evaluated using Eq. (7). As shown in Table 3, owing to the small ε value, the error formed after each contour was approximated to a fitted shape was also small, and the fitting accuracy was very high.

$$accuracy=|contour_area–region_area| \,/\, region_area \tag{7}$$

where *contour_area* represents the area of the contour of each region and *region_area* represents the area of the fitted shape of each region.

Table 4 Error of contour fitting for each region when a structured representation is used

Base station coordinates	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6	Region 7
(125, 300)	0.01	0.02	0	0.02	0.01	0.01	0.01
(250, 350)	0.02	0	0.01	0.01	/	/	/
(400, 100)	0	0.01	0	0.01	0.01	0.01	0.01

In addition, errors may occur during the calculation of communication quality within each region. When the contour recognition method for region division is used after threshold division, if the signal strength distribution of the original communication viewshed data is scattered and varies in strength, then after region division, both strong and weak signals inevitably exist within the same region. Taking the average value of the sum of the communication quality that corresponds to each terrain point within the region as the communication quality value of the region, this process does not distinguish between strong and weak signals; thus, the calculated communication quality of the region may be either overestimated or underestimated. The communication quality error calculation equation for each region is given by Eq. (8). For each region, the error of data compression was measured by the average relative error, which is the average of the relative errors of the signal reception quality at various terrain points.

$$Strength_error = \frac{1}{M}\sum_{i=1}^{M}\frac{|S_i - region_quality|}{region_quality}\times 100\% \tag{8}$$

where M denotes the number of terrain points in the region, S_i represents the signal strength value of the i-th of the M terrain points, and *region_quality* represents the parameter for the

structured representation, namely, the regional communication quality.

The average relative error of the communication quality value in each region for base stations at three different locations is shown in Table 5. The accumulated error was calculated using the three different terrain points used in Section 4.1.1, namely, (125, 300), (250, 350) and (400, 100). As shown in Table 5, the average relative communication quality value of each region in the structured representation was very small. Thus, the small average relative error in Table 5 indicates that the structured representation data compression method does not tamper with the essence of the original communication viewshed data.

Table 5 Average relative error of the communication quality values for each region when structured expressed

Base station coordinates	Region 1	Region 2	Region 3	Region 4	Region 5	Region 6	Region 7
(125, 300)	1.4%	1.5%	3.6%	1.1%	1.6%	3%	2.2%
(250, 350)	3.2%	2.1%	2%	3.5%	/	/	/
(400, 100)	2.4%	5%	1.3%	1.8%	2.7%	1.6%	3.2%

Although there was a certain error between the communication viewshed data after structured representation and the original data, the error was small and did not change the content expressed by the original data; thus, the structured representation was loyal to the original data.

4.2.2 Comparative experiment on structured representation compression methods

The structured representation method for communication viewsheds is a lossy data compression method. Traditional communication viewshed data are typically stored in text in the form of discrete points or TIF images. If the size of the terrain being studied is 500×500, then storing the communication viewshed data for one base station requires 1222 K of space. Storage in the form of TIF images occupies 450 K. However, if the data are structured expressed, the amount of data can be significantly reduced. A comparison of the occupied space for storing communication viewshed data in various forms on terrains of different sizes is presented in Table 6. As shown in Table 6, the storage space occupied by text-form communication viewshed data is highly sensitive to the terrain size. As the terrain size increases, the occupied space increased exponentially. However, if the communication viewshed is structured expressed, the storage volume is less affected by the change in terrain size, and the space occupied during storage is significantly reduced.

Table 6 Comparison of space occupied when communication viewshed data on terrains of different sizes are stored in various forms

Terrain size	Number of base stations	Text	TIF	Structured representation
500×500	1	1222K	450K	3K
	250000	291.35G	107.29G	287.67M
800×800	1	4375K	1252K	4K
	640000	2670.29G	764.16G	963.

To verify the effectiveness of the structured representation compression method, the method proposed in this study was compared with the visibility compression method (CID), which was proposed by Mineter. The communication viewshed data were compressed on

eight terrains of various sizes. The compression ratios of the two compression methods on terrains of different sizes are compared in Figure 14. As shown in Figure 14, the compression ratio of the structured representation compression method proposed in this study increases exponentially with increasing terrain size. Because Mineter's compression method (CID) writes multiple MAP data streams into a single BIL file and then applies ZIP compression to the BIL file, its compression ratio for viewshed data depends on the spatial redundancy characteristics of the terrain data. Under similar terrain conditions, the compression ratio remained relatively stable across datasets of different scales. However, if the terrain complexity increases significantly, the viewshed data become more scattered, thereby leading to reduced redundancy. This directly causes a decline in the compression ratio of Mineter's proposed method.

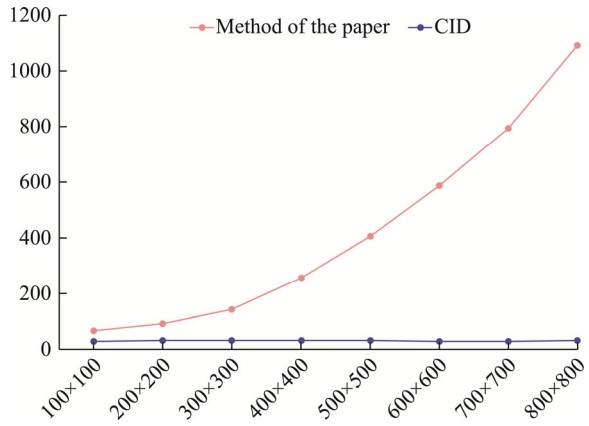


Figure 14 Compression ratios of the two different compression methods for terrains of different sizes

The fundamental distinction between the two methods lies in their compression paradigms: the CID method is lossless, whereas the proposed structured representation method is lossy. Mineter's CID method, which is a lossless compression technique, guarantees the exact reconstruction of the original visibility data. Its compression ratio is inherently limited by the spatial redundancy present in the terrain viewshed data. Under similar terrain conditions, its ratio remains stable, but it decreases significantly when the terrain complexity increases or when the data become more scattered. In contrast, the proposed structured representation method intentionally accepts a controlled, quantified amount of information loss to achieve orders-of-magnitude higher compression. By transforming dense, discrete point data into concise, structured geometric abstractions, we discard precise, point-level signal values in favor of aggregated regional features (e.g., *region_quality* and *region_shape*). This abstraction step results in massive redundancy elimination, which a lossless method cannot achieve. Therefore, the compression ratio of the proposed structured representation method increases exponentially with increasing terrain size.

The CID method preserves 100% fidelity at the cost of lower and data-dependent compression. The proposed structured representation method achieves an exceptionally high compression ratio by sacrificing a small degree of precision while retaining the critical spatial structure and statistical features that are essential for signal coverage analysis. In summary, the CID method is applicable primarily in scenarios that require pixel-perfect visibility

data for precise engineering analysis. In contrast, the proposed structured representation method is targeted mainly at applications in which the spatial distribution, extent, and quality of coverage are sufficient and the data volume, transmission speed, and storage cost are critically constrained, such as large-scale preliminary site planning, real-time network optimization, and mobile GIS applications.

To comprehensively evaluate the compression performance of the proposed structured representation method, in this study, the structured representation compression approach is compared with three established terrain data compression methods: JPEG2000, SPIHT, and Quadtree. As shown in Table 7, in which the compression ratios of these methods are compared on a 500×500 terrain, the structured representation method clearly achieves the most favorable compression results.

Table 7 Compression ratios for each method on a 500×500 terrain

Method	Type	Compression ratio
Proposed (Structured)	lossy	407
JPEG2000	lossy	64
SPIHT	lossy	70
Quadtree	lossy	55

5 Conclusion

To date, various studies on spatial aggregation and distributional characteristics of viewsheds have been conducted. The spatial distribution characteristics of viewsheds are highly important and valuable for their coverage and application. In this study, one branch of viewsheds, namely, communication viewsheds, was considered, and a structured representation method and data compression method for communication viewsheds were proposed. The structured representation method for communication viewsheds includes four steps: acquisition of the communication viewshed, region extraction and fitting, region parameter setting, and single-base-station structured representation. The results show that the structured representation method for communication viewsheds that was proposed in this study can effectively represent the spatial distribution characteristics of signal coverage, thus providing strong technical support for the rapid solution of the base station location problem. Simultaneously, the information storage volume of communication viewsheds was significantly reduced on the basis of structured representation. The experimental results show that this approach is a feasible lossy compression method for viewing information.

However, during the feature extraction step of the structured representation method proposed in this study, a low data distribution density in the region affects the accuracy of feature extraction. In further research, more accurate methods for regional feature extraction should be developed. Furthermore, the proposed method is inherently based on 2.5D terrain representation using a digital elevation model (DEM). Although this is highly effective for analyzing signal propagation over natural terrain (e.g., mountains, hills, and plains) where line-of-sight and terrain diffraction are primary factors, its applicability is likely limited in dense urban environments. In such complex 3D settings, signal propagation is dominated by phenomena that are not captured by a 2.5D surface, such as reflections and multipath effects

between buildings. Therefore, the current method may not accurately represent communication viewsheds in urban canyons. The integration of 3D city models to extend the proposed structured representation framework to urban scenarios could be explored in future work.

Geolocation information statement

Three kinds of terrains were selected for research: mountains, hills and plains. The terrain data for the north district of the Xianlin Campus of Nanjing Normal University were obtained from https://nnu.geodata.cn for applied research.

Availability of data and codes

The data and code that support the findings of this study are available in Zenodo with the identifiers doi: 10.5281/zenodo.15589131 and doi: 10.5281/zenodo.15589412.

Conflicts of interest

No potential conflicts of interest were reported by the authors.

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